



RoadSoS

The AI Companion that turns every bystander into a first responder

Saving lives in the Golden Hour through AI-driven triage,
location-aware emergency dispatch, and voice-first accessibility

CoERS, IIT Madras | Road Safety Hackathon 2026

1.19M

lives lost on roads every year — half within 60 minutes

"Most aren't killed by the crash. They're killed by the wait."

Metric	Global	BIMSTEC
Ambulance response (urban)	8-12 min	20-35 min
Ambulance response (rural)	15-25 min	45-90+ min
Bystander first-aid rate	15-25%	< 5%
Deaths within Golden Hour	~50%	60-70%

Our Solution

A multi-modal AI emergency companion — WhatsApp, voice, SMS, AR —
that orchestrates the **entire chain of survival**.



Speed

One tap, zero friction.
No login walls.



Vernacular

8+ BIMSTEC languages.
Code-mixed speech.



Resilient

Offline-first. SMS fallback.
One bar of signal is enough.

PWA

WhatsApp

IVR Voice

SMS

AR Overlay

How It Works

11 PM, NH-37 Assam. A passing driver, Ravi, stops at a crash scene.



0:00

Ravi taps SOS on PWA — or sends "SOS" on WhatsApp



0:05

GPS locked. AI asks triage questions *in Assamese* (phone locale detected)



0:20

HIGH severity → Ambulance, nearest hospital, police, towing & puncture shops dispatched



0:35

Voice + AR overlay guides Ravi through bleeding control while stabilizing neck



1:10

Scene photos uploaded. Police alerted. Victim's ICE contact called via NFC tag



11:20

Ambulance arrives. ER pre-warned with triage notes. The Golden Hour is intact.

LangGraph

Llama 3.1

FastAPI

Whisper ASR

PostGIS

WhatsApp API

OSRM

Tech & AI Depth

Presentation PWA · WhatsApp · IVR · SMS · AR Camera

AI Orchestration LangGraph Agent · RAG (WHO Protocols) · ASR/TTS · IndicTrans2

Services Hospitals · Ambulances · Police · Towing · Puncture Shops · Showrooms

Data PostgreSQL+PostGIS · pgvector · Redis · Celery

Why RAG, not free generation?

Medical hallucination is unacceptable. Every instruction grounded in vetted WHO + ATLS corpus. Confidence < 0.78 → human dispatcher.

Open-weights LLM

On-prem deploy

Offline-first

14 countries

8 languages

Impact & Roadmap

22-34%

projected mortality reduction

Modelled against WHO trauma curves.
If response time cut by 15 minutes
on BIMSTEC highway corridors.

Open-source SDK

Privacy-by-design

CoERS governance

Stage 1

Jun 2026

PWA MVP, WhatsApp bot, TN seed DB, demo

Pilot A

Q3 2026

2 highway corridors (NH-44, NH-66) with state EMS

Pilot B

Q4 2026

6 Indian states + Sri Lanka + Bhutan via BIMSTEC

Scale

2027

Cross-BIMSTEC rollout, national 112 integration

"Every minute matters. Let's give every road user a fighting chance."



Scan to try RoadSoS



API Documentation

Team RoadSoS · CoERS, IIT Madras · RBG Labs

Road Safety Hackathon 2026 · BIMSTEC Track